(PROJECT 1)

Tamagotchi

Description

A tamagotchi is a digital/physical pet which has 4 states, happy, neutral, angry, and dead, with the main goal of keeping your pet happy. I would build a program that changes and displays the pet's state physically with an ASCII screen for the face and an RGB LED for the mood. The user would feed and pet it using physical buttons which would internally change the pet's mood. These circuits would be surrounded by a custom 3D printed case.

It would look like a wide 3D oval with a rectangular screen on the front, housing the ASCII face, 2 buttons on the body, and the led on the top right, sticking out.

STEM Connection

This project connects to STEM because it requires wide range usage of technology, including software, hardware and CAD. This project would deepen my understanding across these fields by integrating them into a collective system. In addition, this project would be iterated on in order to be functional for the project, this includes debugging code, measuring dimensions and wiring components.

1. Feedback Loops: The process of using output information to adjust a system's performance, ensuring it stays on target.

In a Tamagochi, the inputs are the feed and pet button, while the outputs are the mood indicator light, the screen, and potentially a buzzer. Based on the user's button input, different animations happen on the screen and the mood light brightens. This visual indication tells the user what their action did, providing a feedback loop between the user that interacts with the hardware and the software.

2. Abstraction and Layering: Basically, in order to build systems with billions of components without being overwhelmed, each layer has a simplified interface for the layer above it to interact with, concealing the complexity.

This principle is present in this system because instead of working with transistors and analog circuits to create logic, I use an Arduino which allows me to program the logic gates and the flow of electricity using code. It helps me with signals too, allowing me to get signals from the program and interpret them.

3. Interdependence: Hardware provides the physical, tangible platform, and software provides the instructions.

This relationship is seen in the Tamagochti project. The hardware allows the user to see how their actions affect the program, establishing a connection between the user and the product. Hardware establishes the connection between the user and program via physical input buttons. If the screen didn't change or the indicator light didn't blink, the user could not understand what was happening, preventing usage. The software controls the actions of the hardware, giving the right actions for the hardware to do based on the user's input. If the user presses the feed button, the screen should display the hunger going down or a happy facial expression so the program has functionality.

Demonstration Plan

During the final presentation, I would use the Tamagotchi showing the different reactions the pet has when interacted with, showing initial startup with name choosing, and on the slides I would explain the longer processes like death, hunger, and sadness with actual pictures of the build. If there's time I would go in depth a little about the development process with some flow charts.

Materials List

Already have

- Elegoo Uno R3
- LCD1602 Module
- Female-to-Male Dupont Wires
- 2 Buttons (for testing)
- Tiny Breadboard (for actual use in it)
- Big Breadboard (for initial construction)

Items that must be purchased

- 2 6x6 Tact Switch Tactile Push Button Switch Height 6mm (may not need, if the buttons I have can stick out of the case)

Items available in the shop

- Plastic filament

Software or tools needed

- VScode
- Autodesk Fusion
- 3D printer
- Flush cutters (to clean the print)
- Soldering Iron (if the tiny breadboard doesn't work out)

Cost Estimate

2 6x6 Tact Switch Tactile Push Button Switch Height 6mm - $0.10 x 2 =$0.20

Plastic Filament ~$2

Total = $2.20

Anticipated Challenges

- Though I have a little CAD skills, I never modelled a part to be 3D modelled before, potentially leading to conflict. I would approach this with trial and error though, maybe asking for help.
- I've done the SACA training and a tiny bit of low level breadboarding, but I haven't tried embedded systems and microcontrollers yet so there might be a steep learning curve. But, since I have the parts at home, I can fill some gaps at home with Youtube videos.
- I am juggling AP exam prep, AP Seminar PT2 (tight deadlines), and TSA project polishing, which might impair my ability to work on the project, but I already planned my work windows so I can probably handle it.

Why Does This Project Interests You?

I always wanted to have a Tamagotchi, I just never got one. Like I think it would be cool to have a Tamagotchi keychain that I interact with from time to time. Also, I want to learn

the skills associated with the project. The electronics I am planning to use for the project have been sitting in a box since November, so this project gives the perfect excuse and motivation to learn. More broady, I want to move away from software a little because I've been on it for like a year, so this project is great because it is minimal and a different section programming. Above all, I care about my grade so I will get it done.

Note: This is the project I would most like to do; however, I know it takes longer to do than minesweeper, so build quality might suffer because of my other obligations or I would just be stressed, not quite sure.

(PROJECT 2)

Smart Terrarium

Description

A terrarium is a glass container filled with plants to make a self sustaining ecosystem. In these closed-air ecosystems, humidity is trapped, creating ideal conditions for tropical plants; however, it also might lead to overwatering or mold. To counter this, I would make a smart terrarium system which controls the humidity by opening and closing the lid.

It features a servo to control the opening and closing of the lid, a humidity sensor, an Ascii screen to display humidity levels, and an Arduino(actually Elegoo Uno R3) to connect all of parts.

The lid would be 3D printed with an opening flap that fits a glass jar. The jar would be filled with layers of material with layers being clay balls, mesh, activated charcoal, soil, and the pothos plants. Depending on the jar I find, the size can range from small to medium.

STEM Connection

This connects to STEM by utilizing the science of plants and thermodynamics. It explores how humidity affects an ecosystem, connecting to both plant growth patterns and the effect of heat on water. This also connects to engineering and technology by utilizing a sensor, requiring trial and error, as well as understanding systems by correctly reading and connecting it.

1. Phase changes: Matter exists primarily in 4 states of matter which are solid, liquid, gas, and plasma, with change of one state of matter to another being a phase change.

In the water cycle, the water phase changes in many stages. First, water from soil or a body of water is evaporated by the sun, changing from liquid to gas. Transpiration also contributes to the mass, which is basically water vapor from plants. This cumulative collection of water vapor condenses into clouds becoming liquid water droplets, potentially small solid ice crystals as well. Once heavy enough these fall as precipitate such as rain, snow, sleet or hail. This water then becomes runoff on the land depositing the water back into the body of water or just pools to be evaporated again, in addition to providing water for plants and animals.

A Smart Terrarium showcases this same process, just excluding liquid to solid. The sun and the plants evaporate the water within, changing liquid water to gas water vapor. This is then caught on the glass, condensing into liquid water droplets which fall back to the soil, sustaining the plants' water supply.

2. Microbial ecology - The scientific study of how microorganisms interact with each other, their hosts and their environment.

Mold is a microorganism which spreads in warm, humid places making it a significant threat to the pothos plants in the terrarium. A terrarium closes water inside of it and is warmed by the sun, making it a prime mold-growing spot. The "smartness" in the terrarium design prevents this development by regularly controlling humidity to prevent the ideal conditions for mold. This requires research on the conditions mold grows in an environment in order to effectively program the humidifier controller.

3. Feedback - The process of using output information to adjust a system's performance, ensuring it stays on target.

The Smart Terrarium is a feedback control system, with the humidity sensor being the input and the lid mechanism and display being the outputs. When the sensor reads the humidity, the program makes a decision of what to do. If the humidity is too high, the lid opens, if it is too low or the ideal humidity, the lid closes or remains closed. This feedback loop ensures that the internal humidity prevents mold growth.

Demonstration Plan

During the final presentation, I will explain how all of the layers contribute to the ecosystem and the role of humidity in an environment. The class would see all of the different layers, how the physical humidity display changes when I add water, and how the mechanism works because I will plug my computer in the Arduino and activate it

manually. They also might see some stray wiring and the microcontroller out because I don't plan on hiding it.

Materials List

Already have

Terrarium

- Clay Balls
- Soil
- Pothos cuttings
- Chia seeds (maybe)
- Wire mesh
- Glass jar

Electronics

- Small servo motor
- LCD1602 Module
- Female-to-Male Dupont Wires
- Tiny Breadboard (for actual use in it)
- Big Breadboard (for initial planning construction)
- Elegoo Uno R3
- Humidity Sensor

Items that must be purchased

- Activated charcoal

Items available in the shop

- Plastic filament

Software or tools needed

- VScode
- Autodesk Fusion
- 3D printer
- Flush cutters (to clean the print)

Cost Estimate

Plastic Filament - $1.00

Activated Charcoal - $10.98 (ill only use a little tho)

Total Cost - $11.98

Anticipated Challenges

- Sensors may be difficult to implement due to lack of experience and potential sensor accuracy issues, but I can use trial and error to properly adjust it and ask around
- It might be difficult to mount components on to control the electronics, so I might directly embed them into the CAD lid or simply not mount them.
- Based on the jar size, I may have to use chia seeds instead of pothos plants, so I will germinate them once a project is approved
- Again, I still have AP exam prep, AP Seminar PT2 and TSA project polishing, potentially impairing my ability to work on the project, but I am usually good with time crunches.

Why This Project Interests You

Explain why you would be motivated to complete this project.

Passion and curiosity matter.

I've been wanting to make a terrarium for a while now, especially because I lead a terrarium lab, but didn't make one. Also, I have this big fish-tank-sized one, but there's been mold growing so this sensor could be utilized there to tell me when it is too humid. In addition, I haven't properly tinkered with sensors, making this the perfect opportunity to learn with pressure. Finally, once again, I care about my grades.

Note: I would rather do the other two because this one would take the longest and I don't know if I have the time for it.

(PROJECT 3)

Minesweeper App

Description

Minesweeper is a videogame in which the player tries to mark all the bombs in a square grid area on the screen. Squares are marked with numbers indicating how many bombs surround them if they are in the area around a bomb, and are hidden until the player clicks on them. If the player hits a bomb, they lose.

This app would be made with Flask for the backend routing, Jinja for rendering dynamic HTML templates and Javascript for DOM manipulation for real time changes. The game logic would be done with Python and would have a copy of the grid, represented by dictionaries or potentially a board object which would be updated every time the player clicks something.

STEM Connection

This project connects to technology in STEM because it requires a deep understanding of industry standard software tools in order to build a real-world, functional product. This project requires engineering, since I would have to design and build many layers like board generation logic, frontend and backend connections, and (potentially)data persistence.

1. Separation of Concerns: A principle to manage program complexity by breaking each sphere of functionality into different, isolated sections. This is to make it easier to understand code, speeding up debugging and connection creation across the program.

Minesweeper requires many different layers and functionality, including board data, api routes, logic handlers, and user data. If any of these layers break, the game might be unplayable, making separating concerns essential for clear testing that ensures program functionality. In this program, even though it is not a very big program, I plan to separate everything into flask blueprints, making it easier to find code and build a system.

2. DRY (Don't Repeat Yourself): A software guideline that reduces unnecessary code by ensuring that logic, code, or documentation is concise and is not repeated throughout the program.

Minesweeper has many repeating elements like squares and bombs so it is essential to have dedicated algorithms to display, insert data into, and generate different boards. Also,

since I am building a system from scratch, it is important to have concise documentation that follows the DRY principle for an understandable, clear, and full representation of my program. This is important for grading and when developing frontend and backend connections.

3. Hollywood Principle (“Don’t call us, we’ll call you”): A principle where parts of a program only run when they are called. Basically, low-level components await the command to run from a higher-level system, but don't choose when they're called.

In Minesweeper, program actions like clearing a square, starting a game, or saving a score all rely on player action, like clicks. When the player clicks, the data is sent to the logic in the backend which serves as a framework, and decides which functions should run. A low level function example would be a clearSquares function where the backend calls it and passes in the squares to be dug and the frontend displays it. This displays the Hollywood Principle because the function only runs when something tells it to, rather than running on its own.

Demonstration Plan

During the presentation I would "play" a round, demonstrating clearing land, flag placing, landing on a mine, and showing a score board based on the land cleared and, if build time, time played. The class would see it on my computer(or big screen if provided). In the slides I would show the data pipeline, talk about the development process, and specifics on how the board is generated. If there's time, I would talk about implementation challenges.

Materials List

Items that must be purchased

- N/A

Items available in the shop

- N/A

Software or tools needed

- VScode
- Google (for local hosting)

Frameworks/libraries/languages

- Flask
- Jinja
- Javascript
- Python
- SQL (maybe for saving user progress)

Cost Estimate

Total Cost - $0

Anticipated Challenges

Identify at least two challenges for each proposal:

- Dynamic generating of the pattern might be hard, but I will watch some coding minesweeper Youtube videos if I am stuck
- Clearing the empty squares when the player clicks a blank space might be challenging, but I am planning on having the initial rendering adding a value they all share so I can link all the squares
- Again, I am busy with APs and TSA, so I might have less time, but I will plan features before coding them to speed up the process

Why This Project Interests You

I am already familiar with all of the tools I am going to use, reducing stress and uncertainty that could kill my motivation. And, to be honest, I just want to flex to my minesweeper obsessed friend that I am making it, so that's a source of motivation, as well as the aforementioned grades.

Note: This is my second preferred choice, but it is also the fastest. This is my low-risk option to be chosen if the Tamagotchi seems too difficult or time-extensive.